

HARISH YERRAGUNTLA

Full Stack Software Developer

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PROFESSIONAL SUMMARY

Full Stack Developer with 6+ years building production web applications across the full stack — Vue.js frontends, Python backend services, GraphQL APIs, and PostgreSQL data layers. Experienced embedding in cross-functional Agile teams alongside scientists, engineers, and product stakeholders to translate complex domain workflows into robust, maintainable software. Strong focus on clean architecture, well-tested code, and long-term maintainability. Background spans scientific data platforms, materials science research tooling, and enterprise-scale applications.

TECHNICAL SKILLS

Frontend: Vue.js, Vuex, Vue Router, Vuetify, React.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, responsive design

Backend: Python, FastAPI, Django, Flask, Node.js, Express.js, REST APIs, GraphQL, schema-driven API development

Databases: PostgreSQL, MongoDB, MySQL, Redis — query optimization, indexing, schema design, SQLAlchemy, data access layers

Testing: Jest, Vitest, Cypress, pytest — unit, integration, E2E; test-driven development, frontend and backend coverage

Data Visualization: Plotly.js, D3.js, Recharts — violin plots, radar plots, scatter plots, 2D histograms, cross-plot synchronization

Cloud & DevOps: AWS (ECS, Lambda, S3), Azure (AKS, DevOps), Docker, Kubernetes, GitHub Actions, CI/CD, Git, pull request workflows

Architecture: Full stack design, GraphQL schema design, component-driven UI, scalable API patterns, performance optimization

Agile: Sprint planning, stand-ups, backlog refinement, code reviews, cross-functional collaboration, technical design discussions

WORK EXPERIENCE

Full Stack Developer

Jan 2023 – Present

Progress Solutions Inc. • Remote

- Embedded in cross-functional Agile team building data-intensive Vue.js web applications for scientific and materials science research environments — collaborated directly with scientists, engineers, and product stakeholders to translate complex domain workflows into maintainable software
- Developed and maintained Python backend services and GraphQL APIs supporting scientific data queries — schema-driven development with SQLAlchemy data access layers, PostgreSQL optimization, and efficient query patterns for large research datasets
- Built complex scientific data visualizations in Vue.js using Plotly.js — violin plots, radar plots, 2D histograms, scatter plots with lasso selection, and cross-plot synchronization enabling multi-dataset analysis workflows
- Designed and maintained GraphQL schemas for research data queries — enabled precisely scoped data fetching replacing over-fetching REST patterns across scientific analysis modules
- Built reusable Vue.js component library with Vuetify — standardized chart components, form patterns, and layout systems adopted across multiple research platform modules
- Debugged and resolved complex frontend issues including zoom-level scroll sync, duplicate API call prevention, and Vuetify form submission bugs across the Vue.js application
- Wrote unit and integration tests for Vue.js components using Jest and Python services using pytest — maintained high coverage and integrated test suites into CI/CD pipelines
- Participated in sprint planning, backlog refinement, and architecture discussions — contributed to technical design decisions and supported continuous improvement of engineering standards and development workflows

Graduate Research Assistant

Aug 2024 – May 2025

State University of New York, New Paltz • New Paltz, NY

- Built full-stack academic research platform serving 6K+ users — Python FastAPI backend with PostgreSQL, GraphQL API layer, and TypeScript frontend deployed on AWS ECS with GitHub Actions CI/CD

- Collaborated with researchers and faculty to understand domain workflows and translate them into reliable, maintainable software — comfortable working in research environments with technically complex requirements
- Improved PostgreSQL query performance over 40% through indexing, schema normalization, and query builder optimization; wrote comprehensive pytest and Jest test suites covering all critical flows

Full Stack Developer

Aug 2021 – Jul 2023

Harns Technologies • India

- Delivered multi-tenant content platform in Vue.js, Vuex, Node.js, TypeScript, and MongoDB serving approximately 25K daily users — owned front-end and back-end features across the full stack
- Built reusable Vue.js components and Vuex state management patterns for complex multi-user workflows; designed responsive, maintainable UI with clean architecture and cross-browser compatibility
- Developed Python backend services and REST APIs supporting platform data flows; integrated third-party APIs and maintained technical documentation for all services and data models
- Troubleshoot and resolved Vue.js rendering issues, API response inconsistencies, and data access performance problems across the full stack

Frontend Developer

Dec 2020 – Jul 2021

QS Quacquarelli Symonds • India

- Built high-traffic frontend for approximately 2.5M monthly users — React component libraries, GraphQL data fetching, and Webpack bundle optimization improving page load performance
- Collaborated with backend teams on API contract design and data models; contributed Node.js backend support for API integration and data transformation workflows

PROJECTS

FinGuard – Financial Analytics Platform *React · TypeScript · FastAPI · GraphQL · PostgreSQL · SQLAlchemy · Plotly.js*

- Built Python FastAPI backend with GraphQL schema-driven API layer, SQLAlchemy data access, and PostgreSQL optimization — analytics at sub-200ms response times across complex financial computations
- Wrote pytest integration tests for all GraphQL resolvers and Jest unit tests for frontend components; maintained full schema documentation

ClaimSight AI – Enterprise Workflow Platform *Next.js · TypeScript · Node.js · Python · MongoDB · GraphQL*

- Full-stack application with GraphQL API layer, async background processing pipeline, 6-role RBAC, and sub-200ms MongoDB aggregation on 10K+ records
- Contributed to technical design discussions and architecture decisions; wrote unit and integration tests covering all critical workflow stages

EDUCATION

M.S. Computer Science | GPA: 3.81/4.0

Aug 2023 – May 2025

State University of New York, New Paltz

B.Tech. Computer Science

Aug 2017 – May 2021

Lovely Professional University, India